

Research-Backed Problem Definition Document

1.1 Project Overview

Project Name: Saarthi

One-Line Description: A mobile-first, offline-capable AI heritage guide that delivers accurate, contextual, personalized, and multilingual information to tourists while they are physically exploring Indian heritage sites.

Heritage-Tourism Context: India's cultural heritage tourism contributes approximately 56% of the country's total tourism revenue. The Ministry of Tourism reports around 4.29 billion domestic tourist visits and 25.27 million foreign tourist visits in its latest data, with 20.57 million international tourist arrivals and 9.95 million foreign tourist arrivals recorded for 2024 specifically. The global heritage tourism market is valued at approximately USD 624.6 billion (2025) and is projected to reach approximately USD 937 billion by 2033. Despite this enormous scale, the on-site digital interpretation experience for the vast majority of visitors remains fragmented, unreliable, and impersonal.

Target Environment: Indian heritage sites — including centrally protected ASI monuments (~3,685 sites), state-protected monuments, UNESCO World Heritage properties, and lesser-known cultural landmarks across the country.

1.2 Core Problem

Tourists physically exploring heritage sites in India face five interrelated, well-documented problems:

Problem 1 — Fragmented Information

History, tickets, maps, facilities, transport, opening hours, and accessibility information are spread across multiple disconnected platforms. Tourists must juggle separate sources — OTAs for bookings, ASI for monument lists, state tourism sites for local information, Google Maps for navigation, audio guide apps for interpretation, and informal online content for context. No single platform unifies these into a coherent on-site experience. The Ministry of Tourism itself developed the Incredible India Digital Platform specifically to address this fragmentation, acknowledging that tourists need a central travel-information system.

Problem 2 — Limited and Unreliable Information

Official content is often insufficient in depth and coverage. Fewer than half of surveyed visitors considered ASI website information adequate; on-site ASI office information was rated "good quality but insufficient quantity." Simultaneously, commercial and unverified sources — including tour operator websites, social media posts, and informal guides — frequently outrank official sources in search results, increasing the risk of misinformation. Approximately 53% of travelers report receiving

incorrect heritage information, and ~47% report experiencing travel scams including fake packages and misleading transport.

Problem 3 — Weak Contextual Interpretation

Existing platforms and on-site signage typically provide basic facts — names, dates, and brief descriptions — but lack the location-specific historical, architectural, cultural, and community context that transforms a visit into a meaningful experience. Government reports note that many monuments lack basic historical, cultural, and architectural interpretation. Visitors are rarely informed whether a feature is original, restored, reconstructed, or interpretive — a distinction that ICOMOS guidance stresses is essential. The result is that approximately 74% of travelers seek meaningful cultural experiences, but only ~10% report actually receiving them, with a 40–60% failure rate for depth and education at arrival gateways.

Problem 4 — Limited Personalization and Accessibility

Current solutions rarely adapt to individual visitor needs. Systems do not consistently account for differences in language preference, cultural interests, available time, mobility constraints, age, or prior knowledge. Multilingual signs, audio guides, and accessible-route information are inconsistent across sites. Visitors with disabilities face missing ramps, inaccessible toilets, and poor pre-visit accessibility information. Personalized tourism services remain a "developing objective" rather than a standard feature. Foreign visitors knew about only ~two-thirds of less-visited monuments before their visit, and these sites have weaker signage and digital interpretation.

Problem 5 — Connectivity Dependency

Comprehensive digital interpretation, AR experiences, maps, and navigation are rarely available offline. Many popular heritage sites have 4G coverage, but signal is uneven in remote and rural heritage areas and inside large monument complexes. Users report poor data connectivity near several forts and temples. This creates a fundamental mismatch: the moments when tourists most need digital guidance — while physically inside a heritage site — are precisely when connectivity is most unreliable.

1.3 Core Research Gap

Current solutions mainly address individual needs such as discovery, ticketing, mapping, audio tours, or virtual viewing, but rarely combine **accurate, contextual, personalized, multilingual, location-specific, accessible, updated, and offline** information into one unified on-site experience. This gap is explicitly highlighted across government evaluations, academic analyses, and platform reviews. The ecosystem focuses on efficiency and revenue rather than meaning, authenticity, and preservation. Information remains siloed, unverified, or offline-inaccessible; personalization is largely absent; and local stakeholders — guides, artisans, communities — are not systematically integrated or incentivized.

1.4 Quantitative Evidence

Srno.	Metric	Figure	Source / Status
Q1	Domestic tourists using internet as primary information source	~50.1%	Ministry of Tourism study — <i>source verification required for exact citation</i>
Q2	Foreign tourists using internet as primary information source	~73.8%	Ministry of Tourism study — <i>source verification required for exact citation</i>
Q3	Visitors considering ASI website information adequate	<50%	Survey evidence — <i>source verification required for exact citation</i>
Q4	Indian travelers reporting travel scams (transport, fake packages)	~47%	Survey evidence — <i>source verification required for exact citation</i>
Q5	Travelers reporting incorrect heritage information	~53%	Survey evidence — <i>source verification required for exact citation</i>
Q6	Travelers seeking meaningful cultural experiences	~74%	Global/experiential benchmark — <i>source verification required for exact citation</i>
Q7	Travelers actually receiving meaningful experiences	~10%	Global/experiential benchmark — <i>source verification required for exact citation</i>
Q8	Failure rate for depth/education at arrival gateways	40–60%	Global/experiential benchmark — <i>source verification required for exact citation</i>
Q9	Domestic tourist visits (latest reported)	~4.29 billion	Ministry of Tourism / India Tourism Data Compendium
Q10	Foreign tourist visits (latest reported)	~25.27 million	Ministry of Tourism dashboard
Q11	International tourist arrivals (2024)	~20.57 million	Ministry of Tourism
Q12	Foreign tourist arrivals (2024)	~9.95 million	Ministry of Tourism
Q13	Taj Mahal domestic visitors (FY 2024–25)	~6.26 million	ASI visitor statistics
Q14	Taj Mahal foreign visitors (FY 2024–25)	~0.645 million	ASI visitor statistics
Q15	Centrally protected ASI monuments	~3,685 sites	ASI

Q16	Entry fees cited as barrier (domestic)	31.8%	Survey evidence — <i>source verification required for exact citation</i>
Q17	Entry fees cited as barrier (foreign)	23.3%	Survey evidence — <i>source verification required for exact citation</i>
Q18	Lack of time cited as barrier (domestic)	29.9%	Survey evidence — <i>source verification required for exact citation</i>
Q19	Lack of time cited as barrier (foreign)	21.1%	Survey evidence — <i>source verification required for exact citation</i>
Q20	Monument not in itinerary (domestic)	13.8%	Survey evidence — <i>source verification required for exact citation</i>
Q21	Monument not in itinerary (foreign)	12.9%	Survey evidence — <i>source verification required for exact citation</i>
Q22	Global heritage tourism market (2025)	~USD 624.6 billion	Market research — <i>source verification required for exact citation</i>
Q23	Global heritage tourism market (2033 projected)	~USD 937 billion	Market research — <i>source verification required for exact citation</i>
Q24	Cultural heritage share of India's tourism revenue	~56%	Market/industry estimate — <i>source verification required for exact citation</i>
Q25	India smartphone shipments (2025)	~152 million units	Industry data — <i>source verification required for exact citation</i>
Q26	ARCore-compatible Android models (by 2026)	1,200+	Google ARCore documentation

Note: Figures marked "source verification required" are drawn from the project's compiled research and problem statement but have not yet been individually tied to a specific primary publication URL or DOI. They should be verified against the original Ministry of Tourism Data Compendium, ASI annual reports, and the specific survey instruments before formal publication.

1.5 Stakeholders

Stakeholder	Role & Context
Domestic tourists	Families, students, pilgrims visiting ASI/state-protected monuments (Taj Mahal, Hampi, Khajuraho, lesser-known temples/forts); rely on informal guides or unverified online content for interpretation
International tourists	~20+ million annual arrivals on circuits like Delhi–Agra–Jaipur, Kerala, UNESCO sites; seek authentic cultural narratives but encounter fragmented information and commercialized experiences
Local guides	Licensed and informal guides at high-footfall ASI sites; face livelihood pressure due to outdated licensing, "turf wars," and competition from unverified platforms; operate with static scripts and uneven training
ASI / State archaeology departments	Primary custodians of ~3,685 centrally protected monuments and state-protected sites; responsible for preservation but constrained by budgets, overcrowding, and inadequate digital interpretation infrastructure
Site managers / Conservators	Responsible for day-to-day operations, visitor flow, conservation alerts, and facility management at individual monuments
Artisans	Craftspeople near heritage clusters whose livelihoods depend on tourist spending but remain excluded from formal tourism value chains and digital visibility
Local communities	Residents and cultural practitioners near heritage sites; holders of oral histories, living traditions, and intangible cultural heritage that are rarely represented in digital platforms
Tourism businesses	Hotels, transport providers, experience operators, and homestays that form the broader tourism ecosystem around heritage sites

1.6 Existing-System Limitations

System / Platform	What It Does	Documented Limitations
Booking-first OTAs (MakeMyTrip, EaseMyTrip, IRCTC Tourism)	Transactions — hotels, tickets, packages	Optimized for bookings, not site-specific heritage education or personalized exploration paths; virtually no contextual interpretation

Tourism portals (Incredible India, state tourism sites)	Destination discovery, general travel information	Information siloed from on-site interpretation; coverage uneven; updates not synchronized; limited personalization
ASI systems (website, e-ticketing, SmarakSarthi / Adopt a Heritage 2.0)	Monument lists, World Heritage info, e-ticketing for select monuments, basic amenities (toilets, signage, audio guides)	<50% of visitors find website info adequate; on-site info "good but insufficient"; amenity-focused rather than experience/knowledge-focused; coverage limited to ~30 MoUs and major sites
Audio guides (Audio Odigos)	Audio explanations, site maps, images, historical content for 12 important sites; downloadable tour chapters	Limited to selected sites and languages; no real-time or location-triggered adaptation; no personalization; no integration with navigation or facilities
QR-based systems	Connect physical monuments to digital text, photos, videos, audio	Dependent on internet connectivity; content often static and unverified; no location-aware triggering beyond the QR point; no personalization
Google Maps / Virtual platforms (Street View, Google Arts & Culture)	Locations, routes, photographs, virtual views, digital stories, selected virtual tours (e.g., Ajanta–Ellora, "Wonders of India")	Pre-visit focused; no on-site contextual interpretation; no multilingual audio interaction; no accessibility routing; no real-time operational info
AR/VR heritage solutions (Kerala Tourism AR Heritage Walk, government 3D scanning projects)	AR overlays at specific sites (e.g., Sree Padmanabhaswamy Temple); 3D laser scanning, virtual tours, holograms	Require app downloads, strong internet, compatible devices, accurate GPS/camera tracking, and significant battery; tracking fails in complex heritage environments; limited to flagship sites; may oversimplify stories or fail to distinguish original vs. restored elements
Informal guides	On-site narration and navigation	Static scripts, uneven training, legacy licensing; credentials hard to verify; quality weaker at less-visited sites; risk of misinformation and commercial bias
Fragmented content platforms (Wikimedia, YouTube walkers, social media)	Rich stories, photographs, videos	No verification or provenance; no multilingual accessibility; no integration with on-site navigation or booking; contributes to misinformation and scams

1.7 Problem Statement

Tourists physically exploring India's heritage sites lack a single, reliable, location-aware, multilingual, and offline-capable digital platform that delivers verified, contextual, and personalized heritage interpretation at the exact place and time they need it — resulting in widespread misinformation, superficial experiences, exclusion of local communities, and a significant gap between the 74% of travelers who seek meaningful cultural engagement and the ~10% who actually receive it.